Question # 5 of 10 (Start time: 03:37:45 PM, 26 November 2025)

Let S be a set which is not open, then a function f(z) is analytic in _____.

Select the correct option

○	S
●	an open set containing S

Question # 4 of 10 (Start time: 03:37:09 PM, 26 November 2025)

Consider $\lim_{z \to z_0} f(z) = A$ and $\lim_{z \to z_0} g(z) = B$. Then $\lim_{z \to z_0} (f(z)g(z)) =$ _____.

Select the correct option

○	$A \pm B$
●	$A.B$
○	$A - B$
○	$A + B$

Click

MTH632 - Complex Analysis and Differential Geometry (Quiz 3)

Question # 10 of 10 (Start time: 03:39:53 PM, 26 November 2025)

$f(z) = |z|^2$ is differentiable only at _____.

Select the correct option

○	$z = 1$
○	$z = 2$
◉	$z = 0$
○	$z = -1$

MTH632 - Complex Analysis and Differential Geometry (Quiz 3) — Quiz Start Time: 03:36 PM, 26 November 2025

Question # 3 of 10 (Start time: 03:36:47 PM, 26 November 2025) — **Total Marks:** 1

The derivative of f at z_0 i.e. $f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0}$ provided the limit exists. The function f must be defined is a neighbourhood of _____.

Select the correct option

Reload Math Equations

○	z
◉	z_0

Click to Save Answer & Move to Next Question

MTH632 - Complex Analysis and Differential Geometry (Quiz 3)

Question # 1 of 10 (Start time: 03:36:00 PM, 26 November 2025)

By L'Hopital's Rule,

$$\lim_{z \to i} \frac{z^6 + 1}{z^4 - 1} = _____$$

Select the correct option

●	$-\frac{3}{2}$
○	$-\frac{3}{2}i$
○	$\frac{3}{2}$
○	$-\frac{3}{2}i$

MTH632 - Complex Analysis and Differential Geometry (Quiz 3)

Question # 9 of 10 (Start time: 03:39:14 PM, 26 November 2025)

By L'Hopital's Rule,

$$\lim_{z \to i} \frac{z^3 + i}{z - i} = _____$$

Select the correct option

○	3
○	$3i$
○	$-3i$
●	-3

MTH632 - Complex Analysis and Differential Geometry (Quiz 3)

Question # 7 of 10 (Start time: 03:38:37 PM, 26 November 2025)

Let f(z) and g(z) be two analytic functions, then the composition f (g (z)) of these two functions is _____.

Select the correct option

○	not analytic
◉	analytic

MTH632 - Complex Analysis and Differential Geometry (Quiz 3)

Question # 8 of 10 (Start time: 03:38:51 PM, 26 November 2025)

A point at which a function ceases to be analytic is called a _____ point.

Select the correct option

○	regular
○	non-singular
○	non-regular
◉	singular

MTH632 - Complex Analysis and Differential Geometry (Quiz 3)

Question # 6 of 10 (Start time: 03:38:13 PM, 26 November 2025)

If $f(z) = \frac{z^4+1}{z^2+1}$, then $f(z)$ is not analytic at $z =$ _____.

Select the correct option

○	2
◉	i
○	4
○	1